Module 8: Personal Safety at the Project Site

Unit 8.1: Basics of Work Safety

8.1.1: Identify Corporate Policy for Workplace Safety

A Solar PV System is to be installed as per the safety standards, installation and building codes applicable for that particular location. The installer must be well versed with the organization's corporate workplace safety policies.

Common Corporate Workplace Safety Policy

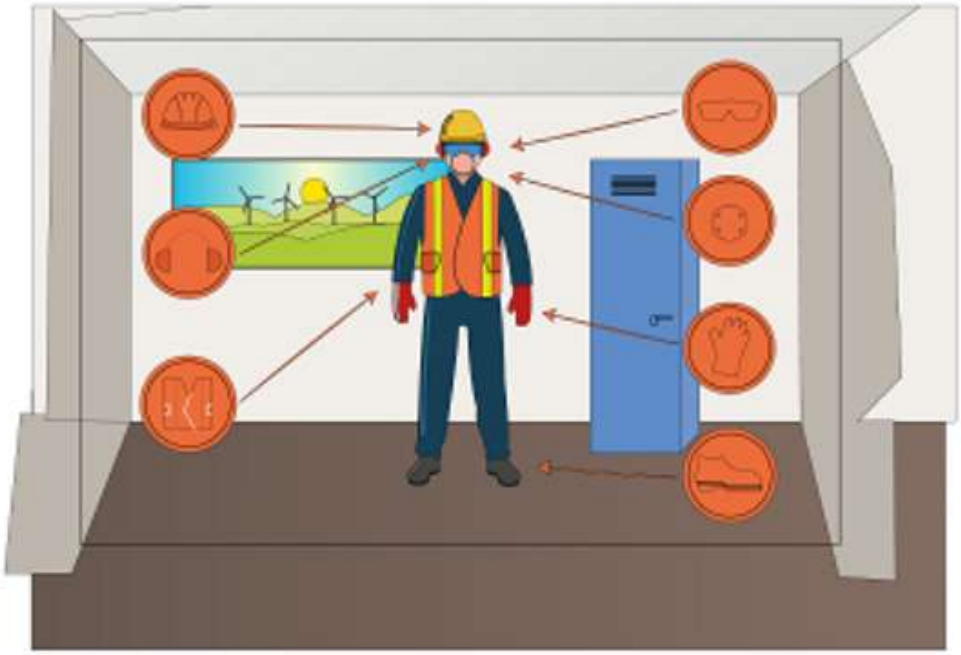
1. Maintaining a safe work area.
2. Safe methods of using tools and equipment
3. Safety of Personnel, which includes knowledge of Personal Protective Equipment (PPE)
4. Awareness of electrical and non-electrical hazards and how to mitigate them

8.1.2: Identify Contact Person when Workplace Safety Policies are Violated



8.1.2: Identify Contact Person when Workplace Safety Policies are Violated

- Site Supervisor or Project Manager
- Health and Safety Officer (HSO) or similar person in organization or location
- Property Owner or Manager
- Human Resources (HR) Department – for large organizations
- A local regulatory body for safety & health administration



8.1.3: Identify Requirements and Create A Safe Work Area Environment

Common Danger Signs on Solar PV Components

PV Array DC Junction Box.
Danger – Contains live parts during day light



PV System – Main AC Isolator

Do Not disconnect DC Plugs and Sockets under load.
Turnoff AC Supply first.

PV Array DC Isolator.
Danger – Contains live parts during daylight

Inverter – Isolate AC and DC Supply before carrying out work.

8.1.3: Identify Requirements and Create A Safe Work Area Environment

General Requirements of a Safe Work Area



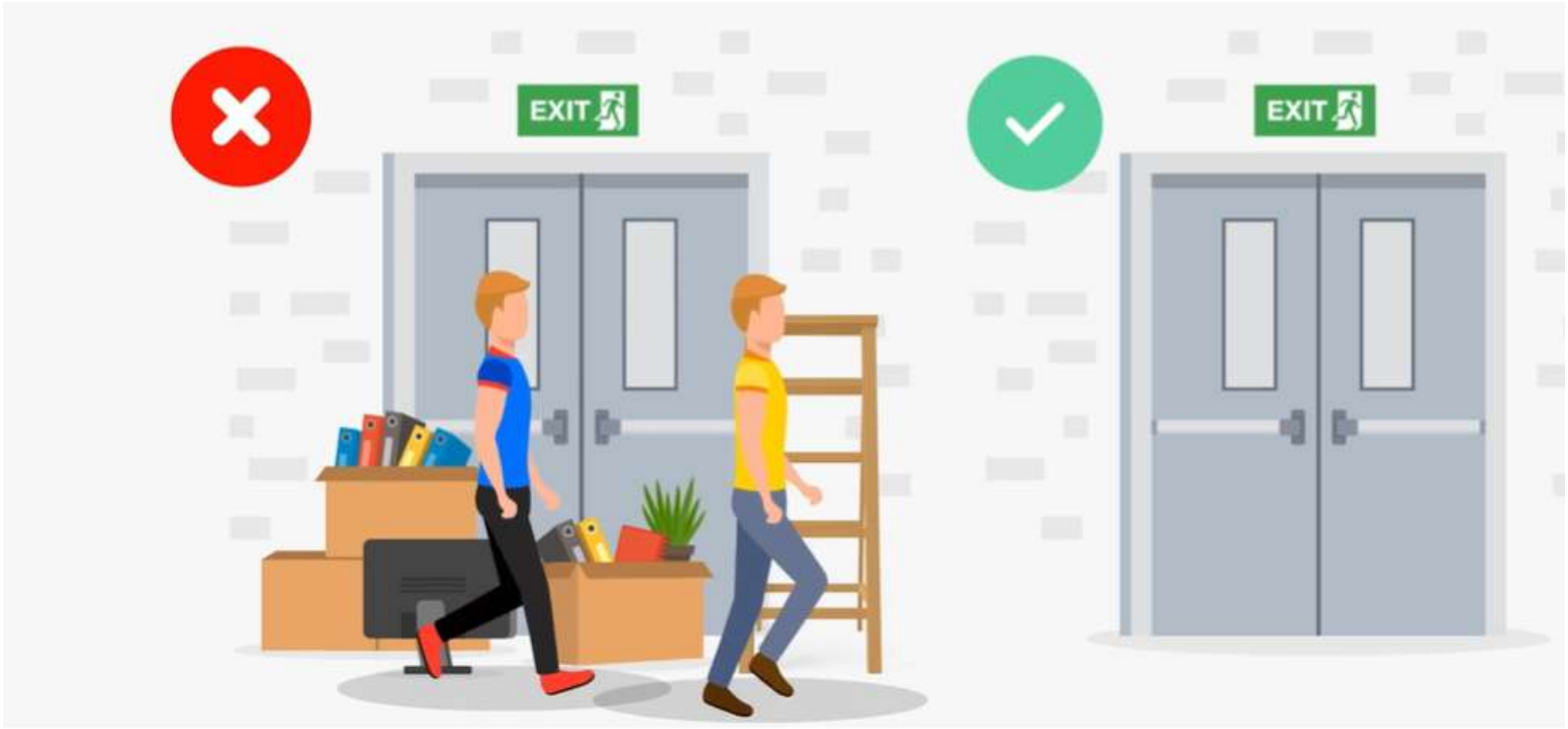
Arya's Safe Practices to Work



Essential Tips on Staying Safe During Solar PV Installations

Ensure Work Area is Clutter-free




EXIT
EXIT

Effective Communication



Essential Tips on Staying Safe During Solar PV Installations

The work site must not be cluttered.

Lighting should be adequate to allow safe working.

The access and exit arrangements should be safe and clear of obstacles.

The tools must not be left unsecured as they may cause obstructions .

Communication arrangements should be adequate

Adequate precautions against sunburns, exhaustion and dehydration

Maintain First Aid Kits to Mitigate Accidents

Unit 8.2: Personal protective equipment (PPE): Importance and use

8.2.1: Identify the Workplace hazards that could cause injury

A **Hazard** is a potential source of harm or adverse effect on a person, property, or the environment. It refers to anything that can cause damage, injury, or adverse health effects.

Personal Protective Equipment (PPE) are important to keep us safe from hazards at workplace.

Some common **causes of hazards** are -

- Intense heat
- Impacts such as cuts from tools, machinery, and materials
- Hazardous chemicals







8.2.1: Identify the Workplace hazards that could cause injury

Type of Hazard	Why it happens
Loss of eye/vision	Using striking tools without eye protection
Puncture wounds	Using a screwdriver with a loose handle which causes the hand to slip
Severed fingers, tendons and arteries	Using the wrong hammer for the job and smashing a finger
Broken bones	Using a small wrench for a big job and bruising a knuckle

8.2.2: Use and maintain Personal Protective Equipment (PPE)



Identify and Use Personal Protective Equipment (PPE)



Personal Protective Equipment (PPE)

1. Eye and Ear Protection



2. Head Protection



3. Foot and Leg Protection



4. Hand and Arm Protection



5. Safety Belt, Body Harness and Overalls



1. Eye and Ear Protection





1a. Eye Protection



Goggles

Why is this necessary?

To Provide Protection From:

- Flying debris, dust and other small particles
- Mechanical impact of tools
- Harmful UV Radiation and strong glare reflected from Solar PV Panels and the sun

1a. Eye Protection



Goggles

Quality Points to consider when choosing the right Googles :

- UV-protective eyewear
- Absorb the shock of a blow
- Glasses with impact-resistant lenses
- Water resistant and slow burning

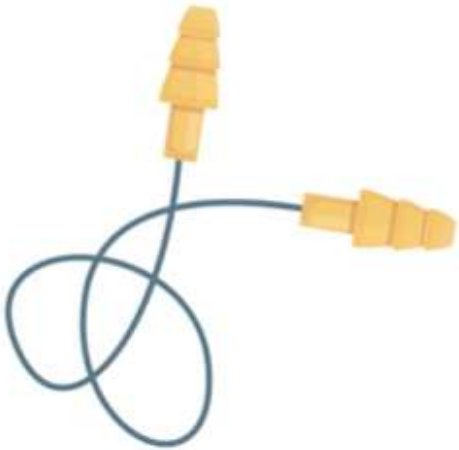
For those who use spectacles, consider

- Goggles that can fit comfortably over corrective eyeglasses without disturbing their alignment.
- Goggles that incorporate corrective lenses mounted behind protective lenses

1b. Ear Protection



Ear Muffs



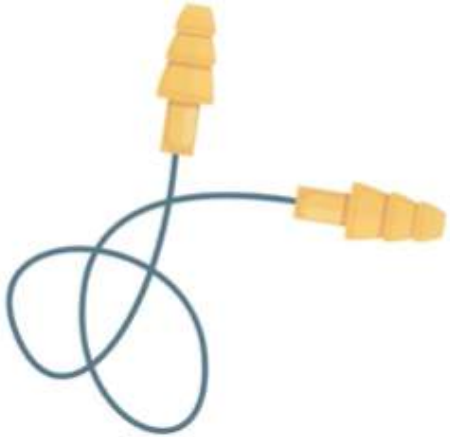
Ear Plugs



1b. Ear Protection



Ear Muffs



Ear Plugs

Why is this necessary?

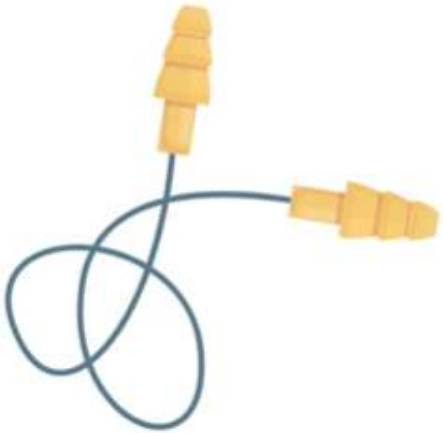
To Provide Protection From:

- Construction Noise of drilling, hammering etc
- Loud noise from heavy machinery such as cranes and transportation
- Environmental noise from nearby areas, such as highways, industries etc.

1b. Ear Protection



Ear Muffs



Ear Plugs

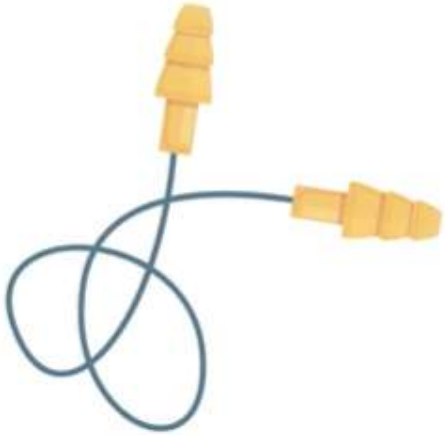
Quality Points to consider when choosing the right Ear Protection :

- Noise Reduction Rating (NRR) of 25-30 dB is typically adequate
- Hypoallergenic and breathable materials to prevent discomfort
- Should be adjustable to fit different head sizes
- Should meet national and international safety standards (e.g., ANSI, OSHA).

1b. Ear Protection



Ear Muffs



Ear Plugs

Quality Points to consider when choosing the right Ear Protection :

Ear Muffs	Ear Plugs
• Suitable for workers who need to wear other headgear such as helmets • More portable and less obtrusive	• Provide a higher level of noise reduction and are easier to put on and take off • Bulkier and may interfere with other PPE

2. Head Protection



Protective Helmets Or Hard Hats

Why is this necessary?

To Provide Protection From:

- Falling objects that might strike on the head
- Fixed objects that might bump heads, such as exposed pipes or beams
- Objects such as exposed electrical conductors

Head Protection Criteria

Protective helmets, or hard hats, should:

- Resist penetration by objects
- Absorb the shock of a blow
- Be water resistant and slow burning

2. Head Protection



Protective Helmets Or Hard Hats

Quality Points to consider when choosing the right Head Protection :

- Resist penetration by objects
- Absorb the shock of a blow
- Be water resistant
- Be resistant or slow to burning
- Should meet national and international safety standards (e.g., ANSI, OSHA - Occupational Safety and Health Administration)

3. Foot and Leg Protection



A pair of ISO 20345:2004 compliant S3 safety boots



Non-Protective Footwear

Why is this necessary?

To Provide Protection From:

- Heavy objects such as barrels or tools that might roll onto or fall on employees' feet.
- Sharp objects such as nails or spikes that might pierce the soles or uppers of ordinary shoes
- Molten metal that might splash on feet or legs
- Hot or wet surfaces
- Slippery surfaces

07:35

3. Foot and Leg Protection



A pair of ISO 20345:2004 compliant S3 safety boots

Quality Points to consider when choosing the right Foot and Leg Protection :

- Should have impact-resistant toes and heat-resistant soles that protect against hot work surfaces common in rooftop installation areas.
- May have metal insoles to protect against puncture wounds
- May be designed to be electrically conductive for use in explosive atmospheres
- May be designed to be electrically nonconductive to protect from workplace electrical hazards
- Should meet national and international safety standards (e.g., ANSI, OSHA)

4. Hand and Arm Protection



Right way: Use protective hand gloves



Wrong way: Use of bare hands

Why is this necessary?

To Provide Protection From:

- Cuts and Abrasions are possible while handling Solar panels, mounting hardware and tools & machinery which often have sharp edges or rough surfaces
- Always a risk of electrical shock, especially during installation and maintenance.
- Chemical of adhesives and cleaning agents

4. Hand and Arm Protection



Protective Hand Gloves

Quality Points to consider when choosing the right Hand and Arm Protection :

- Should have high Material Quality with Cut, Abrasion and Chemical Resistance
- Should be comfortable for Multi-use and tasks such as panel handling, electrical wiring, and general maintenance.
- Be water resistant and slow burning
- Should be electrically insulated for handling electrical connections
- Two kinds of gloves are commonly used: PVC Gloves and Cotton Gloves
- Should meet national and international safety standards (e.g., ANSI, OSHA)

5. Full Body Protection



Body Safety Harness



Safety Jacket





Full Body Suits

5. Full Body Protection





Full Body Suit And Safety Harness

Why is this necessary?

To Provide Protection From:

- Personal protection against falling from high structures
- Enables comfortable working position with protection against slipping or imbalance
- Climbing to a location which is inaccessible from inside the building/household using an anchorage and suspension line

5. Full Body Protection





Full Body Suit And Safety Harness

Quality Points to consider when choosing the right Full Body Protection :

- Material Quality - Should have high quality material resistant to tears, abrasions, punctures as well as flame and UV Rays.
- Should also be insulated from electrical shocks.
- Look for suits made from breathable fabrics for comfort, especially in hot conditions.
- Should allow for a full range of motion without restricting movement, as solar PV installation requires bending, climbing, and reaching.
- Should meet national and international safety standards (e.g., ANSI, OSHA)

Unit 8.3: Identification, Prevention, and Mitigation of Safety Hazards

8.3.1: Identification & Types of Safety Hazards



Everyone wants to avoid hazards – they are dangerous and can be fatal!

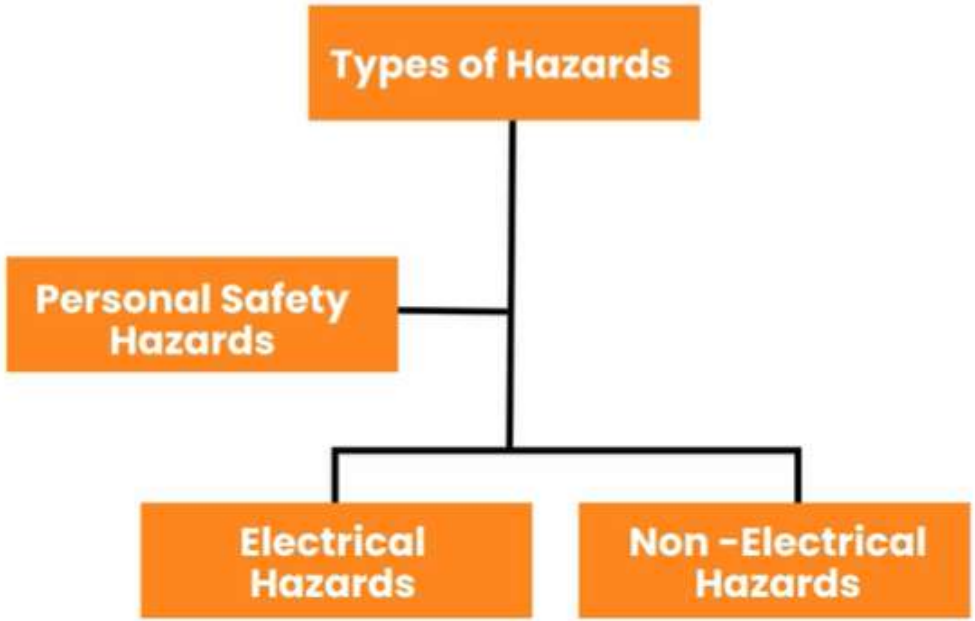
And to mitigate hazards - knowing & identifying the hazards is first step.

Identification & Types of Safety Hazards



Types of Hazards associated with Solar PV Installations











Electrical Hazards



The most common electrical hazards lead to several reactions like

- Electrical shocks
- Burns
- Contraction of muscles
- Injuries from falling after electric shock

Even a very less current (of order mA) is also sufficient to cause damage

Electrical Hazards – Reactions After Shock

1. Perception: tingle, warmth
2. Shock: retain muscle control, reflex may result into injury, burns
3. Severe Shock: lose in muscle control, can't let-go, severe burns, asphyxia
4. Ventricular Fibrillation: may cause death
5. Heart Frozen: temperature of human body rises, death occurs in minutes



Electrical Hazards – Reactions After Shock

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Electrical Hazards – Reactions After Shock

Electrical Burns

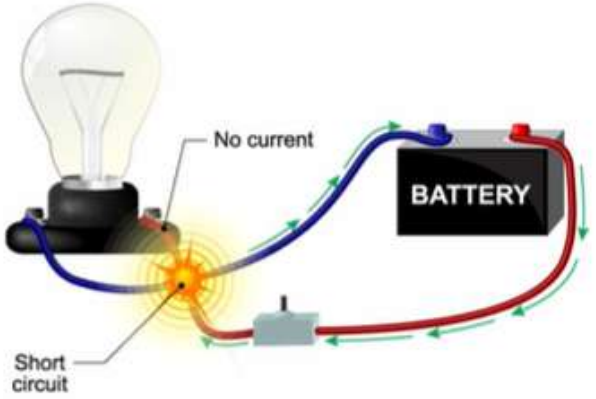
- Shorting the terminals of any typical battery
- Causes severe burns and even death

Acid Burns

- Touching acid with uncovered skin
- Acid contact with human eye can cause blindness

Fire or Gas explosion

- Batteries release Hydrogen gas during their charging
- Flammable gas - can cause gas explosion and fire







Electrical Hazards due to Inverter's heat

When switched on, the surface temperature of inverter is very high and can cause a potential burn hazard.

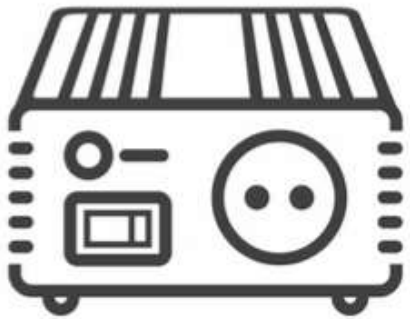
What to do?

- Ensure that the temperature of location should be in the range of -25 to +65 degree F.
- Inverter should be installed at a place where people can't reach or touch frequently.



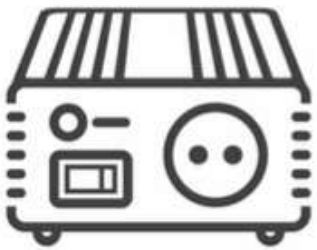


-25 to +65 degree F



Electrical Hazards due to Inverter's high voltage

Solar PV inverter is needed for conversion of DC power from PV array/ modules to AC power.

DC →  → AC



Non-Electrical Hazards

1. Sun Burn & High Sun Exposure

- Sun radiation is very high and no shading exits
- May cause severe sun burns
- Specially in the summer seasons



2. Exposure to Insect & Reptile Bites

- Spiders and other insects often inhabit the junction boxes
- Snakes use the shade built by the PV array
- Ants are also found under PV arrays or near battery storage boxes
- Some wasps build nests in the array framing





3. Skin Injuries such as Cuts and Bumps

- Most of the PV systems & components include metal framing, junction boxes, cables, nuts and bolts, and many cutting tools.
- Their sharp edges can injure, cause cuts and bumps on skin if not cautious or wearing proper PPE



4. Falls and Strains when working at height

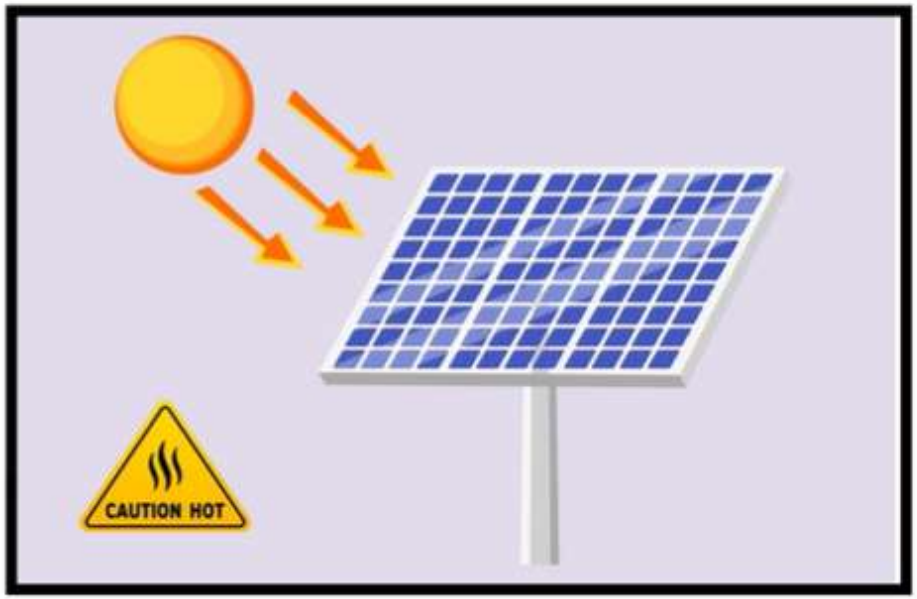
- Walking with heavy loads for installation in remote areas can cause strains

- During strong winds, PV module behaves like a wind-sail and can make you fall from ladder



5. Burns due to high metal temperature

- Metallic surfaces which are left in sun can attain higher temperatures (~ 80 °C) which can cause burns
- Concentrating solar PV system can also cause burn



8.3.2: Prevention, Mitigation & First Aid Practice

- Practice safety measures to prevent hazards.
- Practice first aid in case of hazards.



Prevention, Mitigation & First Aid Measures



1a. General Preventive Measures



1. Identify and understand the company policies required for workplace safety



2. Identify the individual to contact if some accident happens



3. Identify the requirements of safe work



4. Implement the safe workflow procedure



5. Use and maintain Personal Protective Equipment (PPE)



6. Know the location of First Aid box



7. Read and understand the proper usage of tools and equipment



8. Do proper tagging and markings of equipment



9.Add labels or warning signs wherever required for the benefit of technicians as well as customer

1b. Specific Preventive Measures

Component or Procedure-Specific Preventive Measures

1. Safe Handling of Battery

- Use correct posture to lift a battery without causing injury to back and spine.
- Ensure that the battery chemicals do not spill by keeping it straight and horizontal.



2. Safe Lifting & Handling of PV Modules on the Rooftop

- When lifting the pallets of PV modules onto the rooftop, be sure that the roof can support the additional dead weight load of these pallets &that load is distributed evenly.
- Correct equipment to be used for safe lifting and handling of modules.
- Adequately sized project teams are formed for easy handling of modules.
- Lifting equipment is maintained at a safe speed to avoid accident.



2. Safe Lifting & Handling of PV Modules on the Rooftop

- If the modules are lifted manually onto the rooftop, ensure that the path is sanitized and cleared of any physical obstacle



2. Safe Lifting & Handling of PV Modules on the Rooftop

- Use of safety lanyard while working at height is to be ensured always!



3. Installation Contractor Responsibilities

- Reading and understanding all drawing, component and inverter manual prior to installation

- All component required to have switches of position and fuses removed prior to installation



2. Mitigation Measures and First Aid

Despite all precautions and preventions, there are chances of hazards happening.

Hence it is important for all Solar PV Installers to learn & practice important mitigation measures and giving first aid that should be done **immediately after the occurrence of hazards**.







2. Mitigation Measures and First Aid





1. Maintain composure and separate victim from cause of hazard.



2. Wash the injuries and wounds with clean water.





3. Apply First Aid on burns, injuries and wounds.
Different first aid materials are used for different kinds of wounds.





4. Apply Sodium Hydroxide (NaOH) solution when burns are due to battery's acid. Do not wash burns with water.





5. Take the injured person to medical room or a safe place.





6. Look for the Supervisor, Project Manager or any other official contact person in case of hazards.





7. Call the Doctor, Medical Officer or Health and Safety Officer (HSO).



Also know and use safety standards specified such as those by Occupational Safety and Health Administration (OSHA)

Common Types of Fire

A



"Ordinary" combustibles
Paper, wood, rubber, plastics and textiles.

B



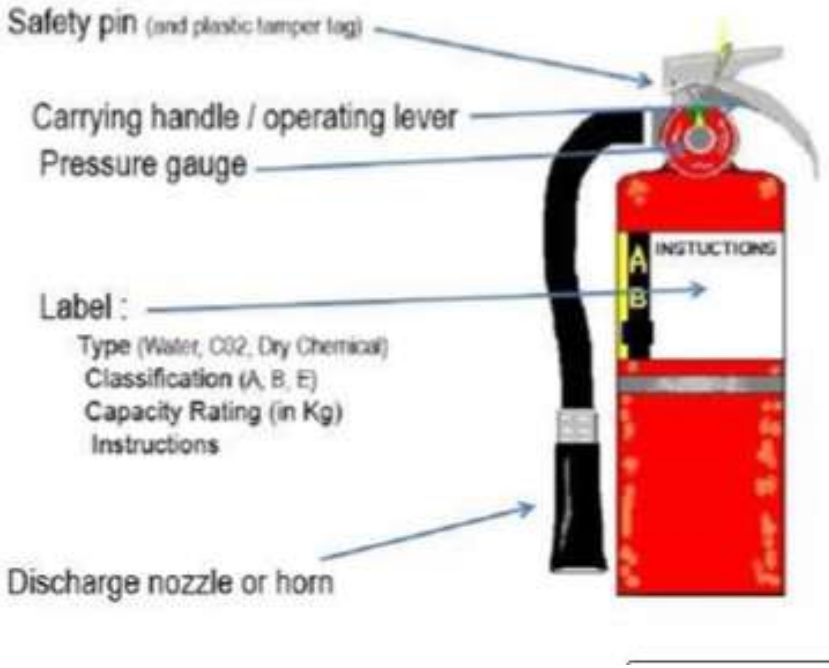
"Flammable liquids"
Oil, petrol, solvents

E



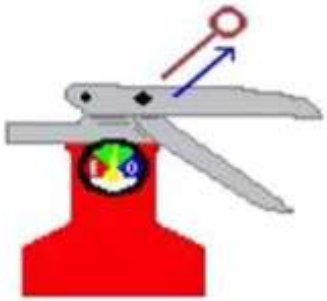
Electrical equipment"
equipment and computers

Common Features of a Fire Extinguisher



Steps to operate Fire Extinguishers

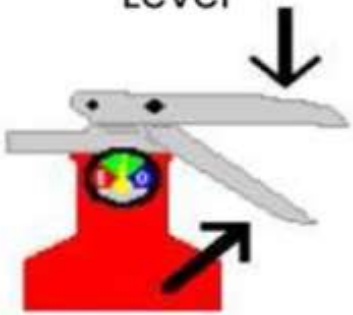
STEP 1 : Pull the Pin



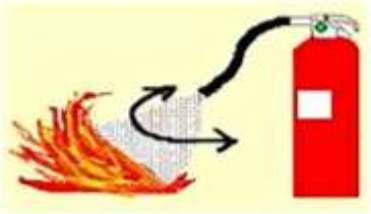
STEP 2 : Aim Low at the Fire



STEP 3 : Squeeze the Lever



STEP 4 : Swipe from Side to Side Slowly



Safety First Always!

7 Steps of Mitigation Measures & Practicing First Aid

1. Maintain composure and separate victim from cause of hazard
2. Wash the injuries/ wounds with clean water
3. Apply First Aid on burns/ injuries/ wounds
4. Apply Sodium Hydroxide (NaOH) solution when burns are due to battery's acid
5. Take the injured person to medical room or a safe place
6. Look for the Supervisor/ Team leader
7. Call the Doctor/ Medical Officer



Unit 8.4: Work Safety and Fall Protection at Height

8.4.1: Work Safety at Heights



- You will know more Safety Best Practices you should take before and while working on sites with heights – such as high-rise buildings and rooftops.
- This will significantly reduce the risk of accidents and injuries.

Work Safety at Heights during Solar PV Installation





As a Suryamitra Solar PV installer, you often spend most time doing installations at heights – such as rooftops of homes and even multi-story high rise buildings. Hence you must ensure **safety practices, fall protection equipment and standard procedures** are in place for you and others in your team before starting to work.

Safety Best Practices for Accessing the Site

Here is a list of Safety Best Practices you should take **before and while** accessing a site at a height to ensure work safety.

1. Perform a thorough **Site Assessment** to identify potential hazards such as weak roof structures, overhead wires, and other obstacles



2. Use correct **Personal Protective Equipment** (PPE) including –

- Full Body Harness and Lanyards correctly anchored
- Helmets with chin straps
- Non-slip Footwear







3. Ensure correct way of using the **full body harness**





4. Use **Ladders and Scaffolding**

- Use only industrial-grade ladders with proper securing mechanisms. Ensure ladders are placed on stable ground and at the correct angle.

- Use scaffolding that is properly erected and inspected. Ensure guardrails, toe boards, and access ladders are in place.

- Always get a second person to securely hold ladders as you climb and use rubber latter mats to prevent the ladder from slipping.





5. Ensure proper **Warning Lines** are made on the rooftop site of work.



6. Ensure proper **lifeline and roof anchors** are installed with lanyard for safety.



7. **Stop work in Bad Weather**

- Never work when it's raining, immediately after rain or in wet or slippery conditions or with wet tools
- Always stop working in bad weather
- PV panels can be blown around by the wind or a storm which can result in you falling or damage to the PV system



8. **Check & Maintain Tools**

- Use right tool for the job
- Inspect all tools before use
- Report to your supervisor any unsafe tool



9. **Careful handling of PV panels**

- Do not apply pressure on PV panels by sitting or stepping on them
- They might break and cause injury, electrical shock or damage to the solar panels
- Never drop anything on the PV panels
- Cover PV panels with an opaque material during wiring to stop or prevent electricity production
- Do not use artificial or magnified light on the PV panels







0:12 / 0:17

10. **Grounding of PV system**

- Make sure your entire PV system is properly and safely earth grounded to prevent electrical shock and injury.





Since at height fall protection is of main importance, you should know the fall protection system & equipment in detail.

Fall Protection Equipment

Equipment used will both protect the fall and absorb some energy of the fall.

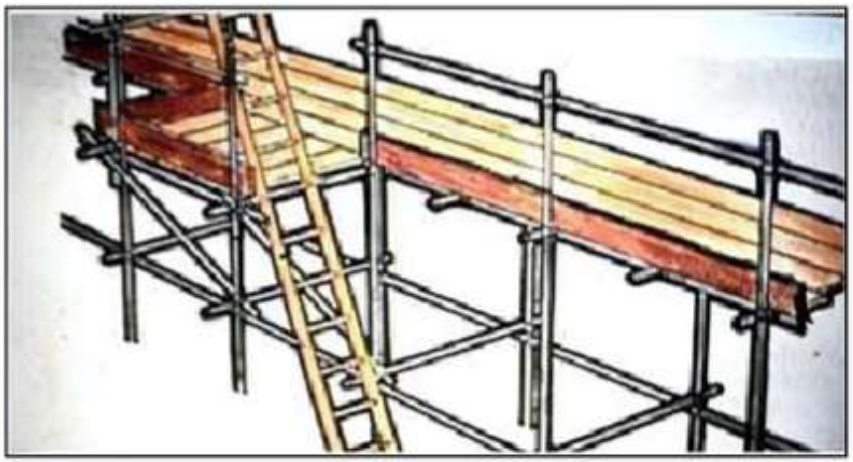
- Full body harness plus lanyard
- Safety nets
- Air bag





Fall Protection System

- Either a guardrail system, safety net system, or personal fall protection system to protect the installer when exposed to a fall of 1.8 m or more from an unprotected side or edge
- Safety nets or airbags should be located as close as possible to the working and walking surface



Guardrails

Use of Safety Signs



Prohibitory Safety Signs

